

Limited-Memory Renewal Contracts: Two-Sided Decomposition and Certified Joint Design

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Abstract. A renewal provider must jointly choose a limited command alphabet and allocate an accepted service obligation under participation caps, realization ceilings, and command-opening charges. We derive an exact price oracle on arbitrary target boxes. Concavity orders the command scores into an increasing prefix and a decreasing suffix: upper-target crossings determine the first side, lower-target crossings determine the second, and a shared peak joins them without exceeding the original budget. The resulting polynomial node oracle yields an adaptive global interval and finite termination at any positive additive tolerance. An exact response-type reduction replaces the number of histories by the number of distinct catalog responses in the search dimension. A separate checker verifies path inequalities, the complete target cover, and the original contract. The worst-case accuracy dependence remains exponential in the number of distinct types; it is not an asserted hardness barrier. Thirty-two synthetic joint-design cases and independent mixed-integer comparisons expose both completed certificates and unresolved gaps. Inherited price-prefix search is stronger on several cases. Canonical contractual allocation, continuous quadratic design, Monge acceleration, budgeted compression, and their proofs remain available under their original assumptions.

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1 Introduction

A service provider may observe many renewal histories while its terminal gateway accepts only a small menu of commands. The provider must choose that menu and allocate its promised service across histories. Participation limits expected service on each history, a separate ceiling limits each realized delivery, and installing a command can incur a fixed charge. Choosing a menu for prescribed branch targets is not the same optimization problem as choosing the menu and those targets jointly. This distinction matters most when the aggregate obligation is below the sum of the participation caps.

We develop an adaptive method for the joint finite-catalog problem. Its central ingredient is an exact price oracle on a box of target intervals. Unlike an unrestricted price problem, a box can force a branch to continue beyond the price-optimal command. Deleting all commands after the priced maximum can therefore destroy the correct upper bound. We show instead that the ordered menu has a two-sided representation: an increasing prefix accounts for upper-target crossings, a decreasing suffix accounts for lower-target crossings, and the common peak accounts for intervals containing it. Two coupled path recurrences optimize both sides at the original command budget, with every selected charge retained.

The resulting oracle yields globally valid bounds for adaptive target-space partitioning. The method returns a feasible contract and an exact interval containing the joint optimum even when stopped before reaching its requested accuracy. A zero-price witness makes the bound exact as a target box contracts. This supplies finite termination at any positive additive tolerance without completing a rectangular target grid. An independently implemented checker verifies all path inequalities, the entire target cover, and the original probability, service, promise, and realization constraints. It does not infer validity from an optimizer’s completion flag.

Repeated histories can be combined when their caps and shortfall costs agree and they admit the same catalog prefix. Their realization ceilings need not be identical. We prove that this reduction preserves the exact joint optimum. The search dimension is consequently the number of distinct response types rather than the number of physical histories. This is an exact structural reduction, not approximate clustering. We also locate the elementary tractable regimes: one response type or a saturated obligation needs one conditional optimization; a fixed command budget admits polynomial catalog enumeration. We do not identify exponential dependence on the number of distinct types as a hardness lower bound.

The computational study evaluates joint, rather than solely conditional, design on a frozen synthetic holdout. Every completed and interrupted run is reported. The comparison includes exhaustive book search, a conventional subset-prefix baseline, independently formulated mixed-integer models, and the prior target grid. Larger repeated-type instances test the exact reduction explicitly; they are not evidence that arbitrary history populations possess a small number of types. The numerical section separates rational global intervals from floating-point solver bounds and from heuristic feasible values.

Relation to established methods. Classical interval quantization and matrix searching provide the foundation for the saturated and conditional path algorithms (Wu, 1991; Nielsen and Nock, 2014; Grønlund et al., 2017). Pricing, constrained paths, and branch-and-bound are established methods (Handler and Zang, 1980; Lawler and Wood, 1966; Lemaréchal, 2001). Recent spatial branch-and-bound methods also exploit target

intervals and locally tight relaxations for separable piecewise-linear optimization (Hübner et al., 2025). Our contribution is not the generic partitioning principle. It is the exact two-sided price decomposition for a shared, charged, cardinality-constrained menu under heterogeneous eligibility, and the root-feasible anchor condition that makes its box bound locally tight. The shared menu couples branches, so the joint objective is not a sum of independently specified piecewise-linear functions. The oracle handles that coupling through two path tables rather than introducing a separate binary decision for every branch–command pair.

The contractual and continuous-design results remain part of the analysis. Canonical allocation connects expected and pathwise institutions; the retained continuous quadratic and Monge results describe the design when command locations are not restricted to a catalog. Fixed-target compression, the full target net, distortion bounds, and charge-aware recovery remain available with their original assumptions. Table 1 distinguishes those antecedents from the present results. The main argument is now organized around joint design, while the companion retains the full preceding theorem chain and its computational evidence.

2 Model and Canonical Allocation

A public branch j is observed with probability $\pi_j > 0$, with $\sum_j \pi_j = 1$. It has a continuation cap $b_j \in (0, 1)$ and a convex nondecreasing intermediate-service cost h_j , with $h_j(0) = 0$. The common terminal reward f is strictly increasing and concave on $[0, 1]$. Caps may repeat; write $\bar{b} = \sum_j \pi_j b_j$. An encoder selects a terminal command from one installed alphabet, and only its symbol reaches the common terminal gateway. The gateway cannot recover the observed branch, an inherited intermediate tier, or an uncharged shared random seed. Thus m distinct commands require $\lceil \log_2 m \rceil$ writable bits; read-only program constants and public-state storage are separate resources. The full institutional mechanism and saturated benchmarks are retained in the companion.

The provider selects a pre-draw tier y_j and a lottery C_j on the installed book, with operating payoff $\mathbb{E}f(C_j) - h_j(y_j)$. The accepted root obligation is $B \in [0, \bar{b}]$. Expected participation restricts the mean obligation, while a realization ceiling restricts every positive-probability draw. With a common risk tolerance $\delta \geq 0$,

$$t_j = y_j + \mathbb{E}C_j \leq b_j, \quad y_j + C_j \leq b_j + \delta \text{ almost surely}, \quad \sum_j \pi_j t_j = B. \quad (1)$$

A book c has at most m distinct levels in $[0, 1]$ and incurs $\sum_{z \in c} \rho(z)$, once per selected level. Let $\mathcal{V}_m^\delta(B; \rho)$ denote maximum operating payoff less this charge. A finite catalog restriction is denoted $\mathcal{V}_{m, \mathcal{A}}^\delta(B; \rho)$. The expectation cap is not replaced by the weaker realization ceiling. Taking $\delta \geq 1$ makes the realization restriction redundant at the canonical optimum below. Taking $\delta = 0$ gives the full pathwise value, even though that institution initially permits draw-specific intermediate service. Thus these are endpoints of one design problem, not three unrelated extensions.

2.1 Eligibility prefixes replace saturated targets

For a strictly ordered book $c = (c_1, \dots, c_s)$ and branch j , define

$$c^{j,\delta} = \{z \in c : z \leq b_j + \delta\}, \quad d_j = \max c^{j,\delta},$$

and let $F_{j,c}^\delta$ be the piecewise-affine interpolant of f on this prefix, including the singleton interpretation. The prefix depends on the cap and risk ceiling, not on the endogenous target.

Theorem 2.1 (Canonical allocation under a realization ceiling). *Suppose f is strictly increasing and concave and h_j is convex, nondecreasing, and zero at zero. A fixed book is feasible exactly when $c_1 \leq \min_j b_j$ and $c_1 \leq B \leq \bar{b}$. Its optimal branch response at target $c_1 \leq t \leq b_j$ is*

$$v_{j,c}^\delta(t) = F_{j,c}^\delta(\min\{t, d_j\}) - h_j((t - d_j)_+). \quad (2)$$

It is attained by $y = (t - d_j)_+$ and an adjacent lottery of mean $\min(t, d_j)$ in the eligible prefix. Consequently

$$\mathcal{V}_m^\delta(B; \rho) = \max_{\substack{|c| \leq m, \ c_1 \leq t_j \leq b_j \\ \sum_j \pi_j t_j = B}} \left\{ \sum_j \pi_j v_{j,c}^\delta(t_j) - \sum_{z \in c} \rho(z) \right\}. \quad (3)$$

The fixed-book inner problem is concave. With zero charges, $\mathcal{V}_m^0(B; 0) = V_m^P(B)$ and $\mathcal{V}_m^\delta(B; 0) = V_m^E(B)$ for $\delta \geq 1$, at every feasible B .

Proof. Every used terminal level satisfies $C \leq b_j + \delta$ because $y \geq 0$. Writing $\mu = \mathbb{E}C$, the payoff is at most $F_{j,c}^\delta(\mu) - h_j(t - \mu)$. Both terms are nondecreasing as μ increases on $[c_1, \min(t, d_j)]$, so this upper bound is maximized at its right endpoint. If $t \leq d_j$, the adjacent lottery has $y = 0$ and all its support lies below $b_j + \delta$. If $t > d_j$, it is the singleton d_j and $y + d_j = t \leq b_j$. Thus the upper bound is attained without relaxing either constraint. The response is concave: its nonnegative interpolation slopes decrease, and its tail has the nonpositive slopes of $-h_j(t - d_j)$. The feasibility conditions follow by using the lowest level and allocating intermediate service; weighted targets fill the entire interval $[c_1, \bar{b}]$.

For pathwise participation with possibly draw-specific Y , all used C satisfy $C \leq b_j$. At fixed $t = \mathbb{E}(Y + C)$, concavity of f on the discrete support and Jensen's inequality for h_j give the same upper bound with $\delta = 0$ and $\mathbb{E}Y = t - \mu$. The canonical policy attains it and is pathwise feasible. Fresh decoder randomization can first be replaced by its conditional mean for each symbol: this increases terminal reward and preserves all the ceilings under the stipulated absence of an uncharged shared seed. Under expected participation the same mean argument has no eligibility truncation; $\delta \geq 1$ includes every level in $[0, 1]$. These observations establish the two endpoint identities and both directions of (3). $\square$

For quadratic primitives (41), chord $[u, v]$ has slope $r - q(u + v)/2$. On branch j its target capacity is

$$[\min\{b_j, v\} - u]_+ \mathbf{1}\{v \leq b_j + \delta\}.$$

Starting at $t_j = c_1$, fill these positive slopes in their common decreasing order, followed by zero-cost

intermediate capacity and the quadratic tails. A water-level sweep over events $\gamma_j(b_j - d_j)_+$ fills the remaining targets. It uses $O(ks + k \log(k + 1))$ rational arithmetic operations, including the certificate

$$\sum_j \pi_j v_{j,c}^\delta(t_j) = \lambda B + \sum_j \pi_j \max_{c_1 \leq t \leq b_j} \{v_{j,c}^\delta(t) - \lambda t\}. \quad (4)$$

The right-hand branch maxima are independently checked at eligible knots, the cap, and the clipped tail stationary point. This multiplier certifies the allocation for that book; it is not by itself a global codebook certificate. The outer problem is solved next.

All canonical arguments continue to hold for branch-specific ceilings $\tau_j \geq b_j$ in place of $b_j + \delta$: the eligibility prefix is then $c \cap [0, \tau_j]$. We write $\mathcal{V}_{m,\mathcal{A}}^\tau(B; \rho)$ for this generalized catalog value. The new budgeted algorithms and numerical formulation use these heterogeneous ceilings. The retained continuous and institutional results use their stated common- δ specialization.

3 Exact Budgeted Compression at Fixed Targets

We first solve alphabet selection without changing the promised branch targets. This is both an independently meaningful design problem and an oracle for joint design. Let $\mathcal{A} = (a_1 < \dots < a_N)$, with charges $\rho_i \geq 0$, and fix $t_j \leq b_j$, $\sum_j \pi_j t_j = B$, and $a_1 \leq \min_j t_j$. Write $\tau_j \geq b_j$ for the realization ceiling; the renewal specialization is $\tau_j = b_j + \delta$. In this section $v_{j,c}$ denotes (2) with this branch-specific ceiling, and $R(c) = \sum_{a_i \in c} \rho_i$.

For two candidate levels $u < v$, define the target-crossing edge and stopping reward

$$A_t(u, v) = \sum_{j: u \leq t_j < v} \pi_j \begin{cases} f(u) + \frac{t_j - u}{v - u} \{f(v) - f(u)\}, & v \leq \tau_j, \\ f(u) - h_j(t_j - u), & v > \tau_j, \end{cases} \quad (5)$$

$$Z_t(u) = \sum_{j: t_j \geq u} \pi_j \{f(u) - h_j(t_j - u)\}. \quad (6)$$

The eligibility test is on v , not on its mean with u . If it fails, a branch uses the lower level and a fixed tier $t_j - u$; averaging an ineligible realization into a feasible mean would not implement the contract. Unlike the priced recurrence, (5) has no price-admissibility restriction on an arc.

Theorem 3.1 (Conditional memory–charge frontier). *For any feasible target vector t , define*

$$F_\ell(t) = \max_{\substack{c \subseteq \mathcal{A}, |c|=\ell \\ c_1 \leq \min_j t_j}} \left\{ \sum_j \pi_j v_{j,c}(t_j) - R(c) \right\}, \quad P_s(t) = \max_{1 \leq \ell \leq s} F_\ell(t).$$

Infeasible exact cardinalities have value $-\infty$. Set $H_1(i) = Z_t(a_i)$ and

$$H_\ell(i) = \max_{v > i} \{A_t(a_i, a_v) - \rho_v + H_{\ell-1}(v)\}. \quad (7)$$

Then $F_\ell(t) = \max_{a_i \leq \min_j t_j} \{H_\ell(i) - \rho_i\}$. All frontiers through budget $M \leq N$ and their feasible policies

are computable in $O((k + M)N^2)$ arithmetic operations and $O(N^2 + MN)$ storage, given constant-time evaluations of f and h_j . Rational quadratic data give polynomial bit complexity. Every policy uses one fixed book, preserves the supplied targets exactly, commits intermediate service before its lottery, and respects every realization ceiling.

Proof. Take a feasible ordered book. Every target belongs to exactly one interval $[u, v)$ between consecutive levels, or to the final interval $[u, \infty)$. In the former case $u \leq t_j \leq b_j \leq \tau_j$, so the lower endpoint is eligible. If v is eligible, the canonical response is its chord at t_j . If v is ineligible, every later level is ineligible, and the highest eligible book level is u ; the response is the second line of (5). A final-interval branch has the response in (6). These intervals partition the branches, including targets equal to a selected level. The book's operating value is therefore the sum of its edge and stopping rewards.

The recurrence enumerates ordered paths with exactly ℓ vertices. It pays every charge once, with the first charge outside the recurrence. Conversely, each such path starting no higher than the smallest target defines a feasible book, and the preceding partition computes its exact value. Maximization proves both directions of the claimed equality. Taking the best exact cardinality gives an at-most budget and does not assume that an unused additional symbol should be purchased. There are $O(N^2)$ edges, each computed by scanning the branches, and $O(MN^2)$ transitions. Each rational path value is a sum of polynomially many input-derived rational terms, so ordinary rational arithmetic and comparison have polynomial bit complexity. $\square$

Corollary 3.2 (Global catalog design at saturation). *If $B = \bar{b}$, then $P_s(b) = \mathcal{V}_{s,\mathcal{A}}^\tau(\bar{b}; \rho)$ for every s . Hence joint catalog design at a saturated promise, including arbitrary nonnegative opening charges and branch-specific realization ceilings, has the complexity in Theorem 3.1, with both k and s variable.*

Proof. Positive weights and $t_j \leq b_j$ imply $\sum_j \pi_j t_j = \bar{b}$ only when every $t_j = b_j$. There is no remaining target optimization. Apply the theorem. $\square$

This corollary is a same-budget global result, not a priced relaxation. At an interior promise, the theorem optimizes the entire alphabet conditional on t ; it does not certify that the selected target vector is jointly optimal. Section EC.12 resolves that target choice at fixed branch count. The faster saturated Monge construction in Appendix E concerns its stronger structural assumptions and is retained rather than replaced.

3.1 Compression, opening costs, and symbol prices

Take any candidate pool U , including a union of supported books. Applying the theorem with $\mathcal{A} = U$ gives the optimal sub-book at every allowed cardinality, not just a sequence of greedy deletions. Applying it on the whole catalog can only improve the conditional value. No nestedness of optimal books is required. In particular, comparing budgets $m, m + 1, \dots, |U|$ gives the complete conditional memory–value frontier and identifies the value purchased by each additional symbol.

If an additional uniform certification fee $\kappa \geq 0$ is imposed, the exact conditional value is

$$\max_{1 \leq \ell \leq M} \{F_\ell(t) - \kappa \ell\}. \quad (8)$$

The upper envelope has finitely many breakpoints, obtained from the computed exact-cardinality values. Chosen cardinalities may be tied, but a largest optimal cardinality is nonincreasing in κ . Indeed, optimality at $\kappa_1 < \kappa_2$ and cardinalities ℓ_1, ℓ_2 gives $(\kappa_2 - \kappa_1)(\ell_1 - \ell_2) \geq 0$. This statement concerns cardinality, not nesting of the selected levels. General nonuniform charges are already included in the recurrence and do not require the uniform-fee interpretation.

The structural advantage over a raw union is that charges enter the optimization itself. A costly level may be discarded even when capacity remains. Nevertheless, an optimal conditional compression can be inferior to a policy with different branch targets. Our returned operational policy therefore compares valid candidates at the *same* budget and keeps the best, including a supplied incumbent. This safeguard never converts a heuristic target portfolio into a global optimality theorem.

4 Response Types and the Complexity Map

Write $P_m(B) = \mathcal{V}_{m, \mathcal{A}}^\tau(B; \rho)$ and $R(c) = \sum_{a_i \in c} \rho_i$. All results in this section concern a fixed finite catalog $\mathcal{A} = (a_1 < \dots < a_N)$. A response type consists of a cap, a shortfall cost function, and an eligible catalog prefix. Let d denote the number of such types; this is distinct from the quadratic reward-curvature parameter q in the continuous-design appendix.

Proposition 4.1 (Exact response-type reduction). *Partition branches into groups with the same b_j , the same h_j on $[0, b_j]$, and the same set $\mathcal{A} \cap [0, \tau_j]$. Replace each group by one branch whose weight is the sum of its weights and whose ceiling is the minimum of its ceilings. The reduced and original joint catalog problems have the same optimum at every feasible B and every budget m . A reduced solution lifts by assigning its target and canonical lottery to every branch in its group.*

Proof. Every book is a subset of $\mathcal{A}$. Group members therefore have exactly the same eligible book and the same concave response v_c . For a group of total weight w , replace its targets by their weighted average $\bar{t}$. This preserves its contribution to the root equality, keeps the average between the common anchor and cap, and gives $wv_c(\bar{t}) \geq \sum_{j \text{ in group}} \pi_j v_c(t_j)$. Neither the book nor its charge changes. Conversely, the common reduced target and its canonical realization satisfy every original ceiling because the eligible catalog prefix is unchanged. The two feasible mappings prove equality of optimal values. $\square$

The common-ceiling homogeneous aggregation argument already appears in the retained continuous analysis. The present statement uses catalog-prefix equivalence and permits distinct numerical ceilings. Equal caps alone do not suffice when either the cost or eligible prefix differs. Forming rational quadratic signatures by binary searching the ordered catalog takes $O(k \log N)$ prefix comparisons, plus grouping and output work. With hashing, grouping uses expected linear dictionary operations; comparison-based sorting instead adds $O(k \log k)$ comparisons. General cost-function equality requires a supplied exact representation or a verified type partition.

Proposition 4.2 (Positive complexity regimes). *For rational quadratic primitives: (i) one response type or $B = \bar{b}$ is solved by one invocation of Theorem 3.1; (ii) for every fixed m , exact joint catalog design is*

polynomial in N , k , and input bit length by enumerating at most $\sum_{s=1}^m \binom{N}{s}$ books and applying the fixed-book allocator; and (iii) the adaptive method below has polynomial work per price per node and a finite additive scheme in dimension $d - 1$.

Proof. With one type the root equality fixes its target. At saturation positive weights force every target to its cap. For a specified book the concave allocation and its supporting-price certificate are polynomial for rational quadratic data, so enumerating the stated finite collection proves (ii). Theorems 5.1 and 6.1 prove (iii). $\square$

Enumeration is an XP bound in m , not a fixed-parameter bound of the form $F(m)N^c$ with a uniform exponent. A hardness assertion with a fixed small m would contradict (ii) unless P equals NP. We establish neither general NP-hardness nor a polynomial-time algorithm when both m and the number of distinct types vary. The additive scheme below remains exponential in d in its worst case. This map distinguishes proved upper bounds from unresolved lower bounds rather than treating the earlier grid's dimension as an intrinsic impossibility.

5 An Exact Two-Sided Price Oracle

After any exact type reduction, retain the index j for a branch and its weight π_j . A target box $X = [\ell, u]$ satisfies $0 \leq \ell_j \leq u_j \leq b_j$ and intersects $\sum_j \pi_j t_j = B$. Only anchors in

$$I_X = \left\{ i : a_i \leq \min_j u_j, \quad \sum_j \pi_j \max\{\ell_j, a_i\} \leq B \right\} \quad (9)$$

can meet the equality inside this box. Because $B \leq \sum_j \pi_j u_j$, these conditions are also sufficient for a fixed book's box feasibility. They depend on its lowest level, not on its remaining levels.

Let $g(x) = f(x) - \lambda x$ at a fixed real price λ . For a book whose first index lies in I_X , relax only the aggregate equality and define

$$S_X(\lambda) = \max_{\substack{\emptyset \neq c \subseteq \mathcal{A}, |c| \leq m \\ c_1 = a_i, i \in I_X}} \left\{ \sum_j \pi_j \max_{\max\{\ell_j, c_1\} \leq t \leq u_j} [v_{j,c}(t) - \lambda t] - R(c) \right\}, \quad (10)$$

$$U_X(\lambda) = \lambda B + S_X(\lambda). \quad (11)$$

Every $U_X(\lambda)$ bounds the best original-budget joint design in X . No zero-duality-gap assumption is made. For a general cost define the clipped scalar support

$$\Phi_j(s, t; \lambda) = \max_{s \leq y \leq t} \{-h_j(y) - \lambda y\}.$$

For quadratic costs it is obtained by clipping $-\lambda/\gamma_j$ to $[s, t]$ when $\gamma_j > 0$; for zero curvature choose the endpoint with the better linear value. Write $\psi_j(z) = \Phi_j(\max\{0, \ell_j - z\}, u_j - z; \lambda)$ whenever $z \leq u_j$.

5.1 The increasing and decreasing sides

For $x < z$ define an upper-crossing edge and its terminal value by

$$A^+(x, z) = \sum_{j: x \leq u_j < z} \pi_j \begin{cases} g(x) + \frac{u_j - x}{z - x} [g(z) - g(x)], & z \leq \tau_j, \\ g(x) + \psi_j(x), & z > \tau_j, \end{cases} \quad (12)$$

$$T^+(x) = \sum_{j: u_j \geq x} \pi_j [g(x) + \psi_j(x)]. \quad (13)$$

When $\lambda > 0$ also define a lower-crossing edge, a lower terminal value, and a peak value:

$$A^-(x, z) = \sum_{j: x \leq \ell_j < z} \pi_j \begin{cases} g(x) + \frac{\ell_j - x}{z - x} [g(z) - g(x)], & z \leq \tau_j, \\ g(x) - h_j(\ell_j - x) - \lambda(\ell_j - x), & z > \tau_j, \end{cases} \quad (14)$$

$$T^-(x) = \sum_{j: \ell_j \geq x} \pi_j [g(x) - h_j(\ell_j - x) - \lambda(\ell_j - x)], \quad (15)$$

$$H(x) = \sum_{j: \ell_j < x \leq u_j} \pi_j g(x). \quad (16)$$

Only increasing-score arcs use A^+ , and only decreasing-score arcs use A^- . Equal-score arcs may be assigned to either side. Set missing forward states to $-\infty$, and compute

$$\alpha_1(i) = \begin{cases} -\rho_i, & i \in I_X, \\ -\infty, & \text{otherwise,} \end{cases} \quad (17)$$

$$\alpha_r(v) = -\rho_v + \max_{\substack{i < v \\ g(a_i) \leq g(a_v)}} [\alpha_{r-1}(i) + A^+(a_i, a_v)], \quad 2 \leq r \leq m.$$

This table pays the charge for its last level. For positive prices a backward table excludes the charge for its first level:

$$\beta_1(i) = T^-(a_i), \quad (18)$$

$$\beta_s(i) = \max \left\{ T^-(a_i), \max_{\substack{v > i \\ g(a_v) \leq g(a_i)}} [A^-(a_i, a_v) - \rho_v + \beta_{s-1}(v)] \right\}, \quad 2 \leq s \leq m.$$

Theorem 5.1 (Two-sided box-price decomposition). *If I_X is nonempty, the exact support in (10) is*

$$S_X(\lambda) = \begin{cases} \max_{1 \leq r \leq m, i} \{\alpha_r(i) + T^+(a_i)\}, & \lambda \leq 0, \\ \max_{1 \leq r \leq m, i} \{\alpha_r(i) + H(a_i) + \beta_{m-r+1}(i)\}, & \lambda > 0. \end{cases} \quad (19)$$

The maximizing book and scalar target supports are recoverable. Computing one price bound takes $O((d+m)N^2)$ arithmetic operations and $O(N^2 + mN)$ storage, where d is the number of branches after type

reduction. General costs additionally require $O(dN^2)$ clipped scalar-oracle calls. Rational quadratic data and a rational price give an exact rational bound in polynomial bit complexity for each node.

Proof. Concavity of g implies that its values on any ordered book increase to a peak and then decrease; a tie can be placed on either side. For $\lambda \leq 0$, strict increase of f makes all scores strictly increasing, so the peak is the final level. Consider first a positive price and a chosen peak p .

If $u_j < p$, exactly one increasing edge $[x, z)$ crosses u_j . If z is eligible, the interpolant of g is nondecreasing through this edge and is maximized on the permitted target interval at u_j . No later segment below u_j exists. If z is ineligible, every later book level is ineligible, so x is the last usable level and clipped shortfall maximization gives $g(x) + \psi_j(x)$. This proves the upper-crossing formula. Its clipping is essential when $\ell_j > x$.

If $\ell_j < p \leq u_j$, the peak is eligible because $p \leq u_j \leq b_j \leq \tau_j$. It attains $g(p)$ at zero shortfall. All eligible chord values are at most $g(p)$, and positive-price shortfall cannot improve them. This proves the middle term.

If $\ell_j \geq p$, the interpolant on the decreasing side is nonincreasing. Each shortfall tail is also nonincreasing because h_j is nondecreasing and $\lambda > 0$. The smallest permissible target, ℓ_j , therefore maximizes the branch's priced payoff. Its crossing edge gives (14); when the next level is ineligible it uses the singleton plus the required shortfall. Targets beyond the last level give (15). Equality at the peak belongs to this third group, preventing double counting.

These three groups partition the branches. Upper crossings before the peak, the peak, and lower crossings after the peak thus decompose the exact priced value of every book, not merely a dominant subset of books. Forward states enumerate all increasing prefixes starting at a feasible anchor. Backward states enumerate all decreasing suffixes, with an optional stop. Gluing a prefix of r levels to a suffix of at most $m - r + 1$ levels counts the shared peak once, counts its charge once, and uses at most m levels. Every feasible ordered book has such a representation, and every represented path is a feasible book for the box. This proves the positive-price formula.

For nonpositive prices, the best point on an eligible increasing chord up to the cap is the cap. Where the next level is ineligible, the last singleton plus clipped shortfall gives the scalar optimum; the same argument gives the final tail. Hence (12) and (13) partition all branches, proving the first case.

The edge construction scans at most d branches for each ordered pair. Each table has mN states and at most N transitions per state. Backpointers reconstruct the book, and scalar maximization reconstructs its supports. For quadratic rational inputs each edge uses a polynomial number of rational operations; each path sums polynomially many terms. Their numerator and denominator lengths remain polynomial in the input, price, and box-endpoint bit lengths. This is a per-node bit-complexity statement, not a bound independent of search depth. $\square$

Why a second side is necessary. Take equal weights, targets and caps $(1/5, 1/2, 4/5)$, unit shortfall curvature, unit ceilings, catalog $\{1/5, 1/2, 4/5\}$, zero charges, budget three, and reward $f(x) = 2x - x^2/2$. At price $3/2$, the complete book has scores $(2/25, 1/8, 2/25)$. On the singleton target box its exact joint value is $169/200$. Restricting to nondecreasing-score books gives only $83/100$, a shortfall of $3/200$. Thus

the unrestricted deletion argument in Theorem A.1 cannot be reused on a target box. The new recurrence repairs that structural obstruction rather than placing the old one-sided oracle inside a generic partitioning procedure.

6 Adaptive Joint Design and Verifiable Global Intervals

Let $\Theta = \{t \in [a_1, b] : \sum_j \pi_j t_j = B\}$. A box can be reduced without losing any point of its intersection with Θ . With $s_\ell = \sum_j \pi_j \ell_j$ and $s_u = \sum_j \pi_j u_j$, its exact coordinate projections are

$$\ell'_j = \max \left\{ \ell_j, \frac{B - s_u + \pi_j u_j}{\pi_j} \right\}, \quad u'_j = \min \left\{ u_j, \frac{B - s_\ell + \pi_j \ell_j}{\pi_j} \right\}. \quad (20)$$

The box is discarded only if $B \notin [s_\ell, s_u]$ or an interval is empty. Formula (20) follows by allocating all other coordinates to their upper or lower endpoints; continuous interval filling proves attainability of each projected endpoint.

At each retained box compute $U_X = \min_{\lambda \in \Lambda} U_X(\lambda)$ for any finite price set containing zero. Every recovered book has an anchor in I_X and hence has a feasible allocation satisfying the original equality. Optimize that book's targets exactly, either within X or over the entire original target region, and update the feasible incumbent L . The implementation uses global fixed-book allocation and caches repeated books. This changes neither the original budget nor the charge schedule. Root price bisection is a choice of bound-strengthening prices, not an assertion of zero duality gap or a prerequisite for validity.

Retain the leaves of a binary cover of Θ , select a leaf with largest upper bound, and bisect one independent target coordinate at its midpoint. Apply (20) to both children. Their intersections with Θ cover their parent's intersection; the common boundary may overlap. A dependent coordinate j_0 is recovered through the equality. Use as splitting coordinate the longest weighted side $\pi_j(u_j - \ell_j)$ among $j \neq j_0$. The code chooses a largest-weight coordinate as j_0 ; this choice is not needed for validity.

Theorem 6.1 (Adaptive original-budget certificate and finite accuracy). *Assume f and all h_j are Lipschitz on their relevant compact intervals and put $L_v = \max\{L_f, \max_j L_{h,j}\}$. Use the exact oracle in Theorem 5.1, include price zero at every node, and evaluate the returned zero-price book by a feasible exact fixed-book allocation. At every interruption,*

$$L \leq P_m(B) \leq U = \max\{L, \max_{X \text{ a leaf}} U_X\}. \quad (21)$$

For every $\varepsilon > 0$, best-upper-bound selection and longest-weighted-side bisection terminate with $U - L \leq \varepsilon$. If $d = 1$ the root is a singleton and is exact. For fixed $d > 1$, a conservative full-tree bound is $2^{H+1} - 1$ nodes, where

$$H = \sum_{j \neq j_0} \max \left\{ 0, \left\lceil \log_2 \frac{2L_v(d-1)w_j^0}{\varepsilon} \right\rceil \right\}, \quad w_j^0 = \pi_j(u_j^0 - \ell_j^0). \quad (22)$$

A zero width contributes zero. With $L_v = 0$, one root evaluation suffices. The bound is polynomial in numerical $1/\varepsilon$ for fixed d , and exponential in d in general; adaptive pruning can finish before this worst-case

cover is constructed.

Proof. A feasible policy supplies the lower bound. Weak duality and the complete leaf cover supply the upper bound, irrespective of the search order or the quality of the incumbent. Thus pruning is valid whenever a leaf bound is at most $L + \varepsilon$, and retaining such leaves in a certificate is harmless.

For local tightness, let c^0 maximize $S_X(0)$ and let $\hat{t}$ be its independently maximizing box targets. Condition (9) gives some equality-feasible t^0 for that same book in X . The canonical response is continuous and piecewise concave with slopes bounded in absolute value by L_v : its chord slopes are bounded by L_f and its shortfall slopes by $L_{h,j}$. Charges cancel for the same book. Therefore

$$U_X \leq U_X(0) \leq \sum_j \pi_j v_{j,c^0}(t_j^0) - R(c^0) + L_v \sum_j \pi_j (u_j - \ell_j) \leq L + L_v \sum_j \pi_j (u_j - \ell_j). \quad (23)$$

An exact global fixed-book allocation can only improve the displayed feasible value. The inequality does not require a uniform bound on a recovery-compression penalty and does not mix two books.

After coordinate projection, the dependent weighted width is no larger than the sum of the other weighted widths. To see this, choose equality-feasible points attaining its lower and upper endpoints and subtract their equalities. Thus the last sum in (23) is at most twice the sum of the independent weighted widths. No node needs further splitting once each independent width is at most $\varepsilon/[2L_v(d-1)]$. Before this occurs, a longest-width split halves a width above that threshold; projection cannot increase any width. Along a path, coordinate j can be split at most its summand in (22) before falling below the threshold. Hence every path needing refinement has at most H splits. The complete binary tree bound follows. At a singleton box the same calculation gives equality of the box upper bound and the fixed-target optimum. This also proves the one-type case. $\square$

The result replaces completion of a full target net by certificate-directed refinement. It does not promise that refinement is inexpensive for every input. It supplies a structural polynomial node oracle and a meaningful interval when it is not. An absolute tolerance is measured in payoff units. Our quadratic experiments normalize target levels to $[0, 1]$ and use $r = 2$, $q = 1$; they report both the absolute gap and its ratio to $r + \max_j \gamma_j/2 + \sum_{i=1}^m \rho_{(i)}$, where the charges are sorted decreasingly. This positive declared scale remains meaningful when net value is close to zero. Rescaling all rewards and charges requires rescaling the requested tolerance.

6.1 Independent checking and physical implementation

A certificate contains the original inputs, the exact type partition, an original-space lottery, every binary split, the authoritative leaves, and one upper-bounding price table per node. It need not contain every price tried. The checker verifies anchor bounds, every reachable forward Bellman inequality, every backward stopping and transition inequality, and the final maximum in (19). Upper tables need not equal the optimizer's tables; satisfying these inequalities is enough to dominate every represented path. It separately reconstructs both projected children at every split and verifies that no leaf is missing. Finally, it checks probabilities, selected charges, pre-draw service, the exact root promise, and supportwise realization limits without using

the canonical response formula. The certificate therefore certifies all target regions, not just the sampled incumbents. Its storage is $O(TmN + Td)$ rational entries for T recorded nodes, apart from the policy and input data; the checker recomputes edge values and has polynomial work per recorded node.

The terminal alphabet is only one physical resource. A type-reduced implementation additionally stores a history-to-type encoder, a target and lottery table, command constants, and probability representations. Arbitrary rational probabilities may require variable-length random-bit generation or a separate finite-precision error analysis; we do not count them as free writable memory. Offline certificates have the size just stated and are not assumed to fit in the terminal gateway. The total deployment cost is consequently not identified with $\lceil \log_2 m \rceil$ bits. This separation is essential to interpreting the repeated-type computation as joint design of a restricted interface, rather than a claim about full-system memory or calibrated field performance.

7 Local Distortion and Same-Budget Certificates

The target net provides a global guarantee when fully enumerated. A complementary certificate can be evaluated with just one conditional dynamic program. Let

$$t_j^0 = \min\{b_j, z\}, \quad \sum_j \pi_j t_j^0 = B, \quad W_0 = \sum_j \pi_j f(t_j^0).$$

At $B = \bar{b}$ take $t^0 = b$. The water level is otherwise found by sorting caps. Let $\rho_* = \min_{a_i \leq a_*} \rho_i$, where $a_* = \min\{B, \min_j b_j\}$. Feasibility guarantees that this set is nonempty.

Theorem 7.1 (Distortion–charge certificate). *For every symbol budget $s \geq 1$, define*

$$C_s = \min_{\substack{c \subseteq \mathcal{A}, |c| \leq s \\ c_1 \leq \min_j t_j^0}} \left[\sum_j \pi_j \{f(t_j^0) - v_{j,c}(t_j^0)\} + R(c) \right].$$

The conditional dynamic program computes $C_s = W_0 - P_s(t^0)$ exactly, and

$$P_s(t^0) \leq \mathcal{V}_{s,\mathcal{A}}^\tau(B; \rho) \leq W_0 - \rho_*. \quad (24)$$

Thus $C_s - \rho_$ is an explicit original-budget optimality-gap certificate. Any other valid upper bound for that budget can replace the right-hand side by its minimum with $W_0 - \rho_*$. A larger-budget policy value is not such an upper bound.*

Proof. For any policy, $\mathbb{E}C_j \leq t_j$ and Jensen’s inequality and monotonicity give $\mathbb{E}f(C_j) - h_j(y_j) \leq f(t_j)$. Maximizing the sum of these ideal rewards under the root equality and caps gives t^0 : their marginal rewards are equal on uncapped branches, and weakly larger on capped branches. This remains an optimum when f is not strictly concave. Every feasible nonempty book contains a level at most a_* and pays at least ρ_* . Hence its net value is at most $W_0 - \rho_*$. Conversely, the conditional optimal policy at t^0 is feasible at budget s . Finally, minimizing the displayed distortion plus charge is exactly the conditional objective subtracted from W_0 . $\square$

The certificate remains valid with nonuniform or large fees, because those fees are optimized rather than charged after an uncontrolled expansion. It can still be loose: C_s includes unavoidable finite-memory distortion, the cost of the selected levels, and any loss caused by fixing t^0 . We report its numerical size instead of asserting that it is uniformly small relative to the optimum. A multiplicative guarantee is particularly inappropriate when the optimal net payoff is zero or negative.

7.1 Local errors, adaptive candidates, and constructive rates

For a twice differentiable f , suppose $-f'' \leq M_{uv}$ on $[u, v]$. If an eligible pair brackets a target t , its interpolation error satisfies

$$0 \leq f(t) - \left[f(u) + \frac{t-u}{v-u}(f(v) - f(u)) \right] \leq \frac{M_{uv}}{2}(t-u)(v-t). \quad (25)$$

This follows by comparing f with its chord plus $M_{uv}(t-u)(v-t)/2$, whose difference has nonnegative second derivative and zero endpoint values. If the next selected level is ineligible or no upper level exists, the exact local deficit is

$$f(t) - f(u) + h_j(t-u), \quad (26)$$

not a second-order chord error. Summing these local quantities with the actual selected-level charges yields C_s for the chosen book. A piecewise curvature majorant can also be placed directly in the target-crossing recurrence to minimize a certified distortion bound. It must retain the first-order expression (26) on ineligible edges.

An adaptive candidate pool can include ideal target values and refine intervals that matter for (25)–(26). The resulting certificate is evaluated from the final pool and policy. We make no uniform convergence-rate claim for the greedy midpoint refinement used in the experiments. In particular, adding a level just above an eligibility threshold need not help the affected branch. Candidate selection and the exact optimization over those candidates are distinct tasks.

Proposition 7.2 (Original-budget comparison books). *Let $m \geq 2$, $h = 1/(m-1)$, and suppose the catalog contains the uniform m -point book $G = \{0, h, \dots, 1\}$. With zero charges, the conditional policy at t^0 satisfies*

$$\mathcal{V}_{m,\mathcal{A}}^\tau(B; 0) - P_m(t^0) \leq L_f h + \sum_j \pi_j h_j(h), \quad (27)$$

$$\mathcal{V}_{m,\mathcal{A}}^\tau(B; 0) - P_m(t^0) \leq M h^2 / 8 \quad \text{if } \min_j(\tau_j - b_j) \geq h \text{ and } -f'' \leq M. \quad (28)$$

In (27), $h_j(h)$ may be replaced by $h_j(\min\{h, b_j\})$ if costs are defined only on $[0, b_j]$. The same bounds hold relative to the uncharged continuous m -level optimum. With charges, adding $R(G) - \rho_*$ to the right-hand side gives the corresponding catalog comparison. If the distinct ideal targets are themselves available and number at most m , the uncharged policy attains W_0 and is globally optimal, for any realization headroom.

Proof. At each target, round its terminal level down to the nearest point of G and supply the difference before execution. The difference is at most h , the target and root promise are unchanged, and every realized total

equals the target, hence is no larger than b_j . This proves (27) by comparison with W_0 . Under the headroom assumption the upper neighbor is no higher than $t_j^0 + h \leq b_j + h \leq \tau_j$, so the adjacent unbiased lottery is feasible with no intermediate service. Apply (25) and $(t - u)(v - t) \leq h^2/4$. The conditional optimum on the whole catalog is no worse than either comparison policy. Since W_0 also bounds continuous design, the continuous comparison follows. With fees, the comparison book pays $R(G)$ while the catalog upper bound subtracts ρ_* . Finally, installing the distinct ideal targets realizes them deterministically with no intermediate service, attaining the ideal upper bound. $\square$

These are explicit same-budget guarantees, not claims that uniform catalogs are always efficient. Their approximation parameter is the number of installed symbols, whereas Theorem EC.12.2 controls computational accuracy at a fixed symbol budget. The local certificate and the direct continuous comparisons below distinguish the two effects. In particular, driving a price tolerance to zero cannot remove finite-catalog distortion or the original-budget price gap.

8 Computational Evidence for Joint Design

The primary protocol fixes 24 independent-seed synthetic cases with two, four, or eight input branches; nine or seventeen catalog points; two or three installed commands; and zero or nonnegative heterogeneous charges. Eight additional cases vary the raw history count and response-type count using a 33-point catalog. Levels lie in $[0, 1]$, the reward has $r = 2$ and $q = 1$, and shortfall curvatures are drawn from $\{1, 2, 4, 8\}$. Participation caps, weights, and eligibility ceilings vary independently within the declared rational generator. The protocol was frozen before these runs, not externally preregistered. Every generated case is retained. Larger comparator extensions were added after the primary run and cover the entire declared comparison set, not selected successes.

The proposed method uses eight root price-bisection steps, retains price zero, and uses the same budget and charges in every incumbent and bound. Primary box runs have a 301-node and eight-second soft limit; repeated-type scaling runs have a 101-node limit. Time is checked after completing a node, so some recorded times exceed eight seconds. The only initial incumbent is a fixed-target conditional book at the capped ideal targets, followed by exact allocation for that book. It is not a global certificate. The baseline price-prefix search reimplements the inherited R43 exact completion bound, uses the same price budget, and stops at the same absolute tolerance. A weaker full-catalog prefix relaxation is reported separately, not substituted for this stronger baseline.

Table 2 reports 28 of 32 box runs attaining absolute gap 0.001, versus 32 of 32 for the priced-prefix baseline. The box method has lower measured optimization time in 11 cases, but it is not uniformly faster or more accurate. Median times are 0.093 seconds for boxes and 0.070 seconds for priced prefixes. Four box runs retain unresolved gaps between 0.00217 and 0.01248. These are valid intervals, not achieved-tolerance claims. The median Jensen-minus-incumbent interval is 0.01124; the median final box interval is 0.00018. Every box interval passes the independent structural checker.

The eight smallest holdout cases also have independent all-book joint references. Their incumbent losses are zero in this sample, but the priced bounds need not be exact. This observation is not a guarantee for the

initial target choice or any heuristic portfolio. Repeated-type cases with 128 and 256 histories reduce exactly to four or eight response types; their certificates are lifted and checked on the original history space. The sixteen-type cases have no such large reduction. These rows demonstrate full joint intervals after a stated exact aggregation, not merely the runtime of conditional target compression.

Table 3 tests the complete-grid reference at successively smaller tolerances and with two through five branches. Only the three two-branch grid runs complete within the 1,000-profile limit. The adaptive method completes eleven of twelve corresponding requests; its four-branch, 0.001 run remains interrupted. Exact-reference losses, normalized objective scales, and all counts are provided rather than inferring accuracy from sampled objective agreement. This comparison tests the new coverage mechanism, not a claim that generic spatial partitioning is novel.

The independent mixed-integer formulation retains installation binaries, branch support indicators, probabilities, pre-draw service, the root equality, and every realization implication. Its tangent and secant models are solved on all 24 primary cases with 64 cost segments and separate four-second limits. All 48 solves return optimal numerical status. Table EC.2 gives each case’s runtimes, search nodes, primal–dual gaps, envelope error and bracket width. Negative signed widths at floating roundoff scale are retained in the raw records and are not called exact certificates. The results show that generic MIP and inherited prefix search remain competitive on these catalogs. The methodological claim for the box oracle is exact lower-bound-aware decomposition and independently checkable adaptive joint certificates, not universal computational superiority.

Regression includes 600 exact all-book price comparisons on heterogeneous target boxes, 24 further joint interval comparisons, 2,793 independent finite-lattice checks of constructive net coverage, genuine below-anchor and above-cap clipping, the one-branch one-call case, and eleven rejected certificate mutations. The three-target descending-side example in Section 5 is an exact counterexample to reusing a rising-only oracle on a target box. The coverage checker is structurally independent of the search routine. These are verification tests, not statistical evidence about an operational population.

The complete historical numerical study remains in Section EC.17. Its thirty-five-of-thirty-six portfolio result and safeguarded sample exactness are not used as the present global-algorithm headline. Its inserted-ideal-target continuous example is an oracle candidate-pool sanity check, and its large conditional table is explicitly labeled “targets fixed.” No customer records, estimated installation charges, field calibration, or deployment outcomes are asserted.

9 Conclusion

A target box changes the structure of the priced menu problem: a mandatory lower target can make commands after the priced peak indispensable. An exact increasing–decreasing path decomposition preserves those commands, accounts for their charges, and optimizes the original limited alphabet. Combining that oracle with exact response-type reduction and independently checked target covers yields a joint-design interval at every interruption and a finite additive scheme at every positive tolerance. The worst-case search remains dimension dependent; the empirical comparison retains instances where inherited prefix search

performs better. The broader contractual and continuous-design results remain available under their original assumptions. A calibrated application, general complexity lower bounds, and improved branching or price selection are further research questions, not unreported empirical accomplishments.

Code and data availability. The supplementary source package contains the exact rational solver, an independently implemented certificate checker, frozen instance protocols, all completed and interrupted outputs, numerical MIP comparisons, and build instructions. Current sources and preserved predecessor derivations are separated by revision directory. The README identifies the current article, electronic companion, response, reproducible commands, and the exact source and review commits. No proprietary data are required.

A Eligibility-Prefix Decomposition

The closed-cell formulation establishes exact continuous attainability. We now exploit the ordered interface to avoid enumerating either branch modes or catalog subsets in a price subproblem. The resulting bounds support finite global search and remain valid when that search stops early. This distinction matters: a price relaxation is not, without an additional equality, the original optimum.

A.1 A reusable ordered-interface allocation class

Let $\mathcal{A} = (a_1 < \dots < a_N) \subset [0, 1]$ be a catalog with nonnegative opening charges ρ_i . The common reward f is strictly increasing and concave. Branch j has weight π_j , target cap b_j , nondecreasing convex shortfall cost h_j with $h_j(0) = 0$, and an eligibility threshold $\tau_j \geq b_j$. Its response is (2) with eligible levels $c \cap [0, \tau_j]$. The renewal model has $\tau_j = b_j + \delta$; the theorem below permits distinct thresholds without requiring a common overrun tolerance. This abstract class needs no acceptance or renewal interpretation. It describes any ordered shared interface with a common concave output reward, separable shortfall costs, and these target and support restrictions.

Write $a_* = \min\{B, \min_j b_j\}$. Assume $a_1 \leq a_*$, since otherwise no feasible book exists, and replace m by $\min\{m, N\}$. Restrict books to $c_1 \leq a_*$; this retains exactly those books that can meet the root promise. At a real price λ , define

$$g_\lambda(x) = f(x) - \lambda x, \quad \phi_j(z; \lambda) = \max_{0 \leq y \leq z} \{-h_j(y) - \lambda y\}, \quad (29)$$

$$D(\lambda) = \lambda B + \max_{\substack{\emptyset \neq c \subset \mathcal{A}, \\ |c| \leq m \\ c_1 \leq a_*}} \left[\sum_j \pi_j \max_{c_1 \leq t \leq b_j} \{v_{j,c}(t) - \lambda t\} - \sum_{a_i \in c} \rho_i \right]. \quad (30)$$

Weak duality gives $\mathcal{V}_{m, \mathcal{A}}^\delta(B; \rho) \leq D(\lambda)$ in the renewal specialization. For quadratic shortfall costs, (29) is evaluated at $y = 0$ if $\lambda \geq 0$, and at $y = \min\{z, -\lambda/\gamma_j\}$ otherwise; when $\gamma_j = 0$ and $\lambda < 0$, use $y = z$.

Call an arc $u < v$ admissible when $g_\lambda(v) \geq g_\lambda(u)$. Define its reward and the reward for stopping at u by

$$E_\lambda(u, v) = \sum_{j: u \leq b_j < v} \pi_j \begin{cases} g_\lambda(u) + \frac{b_j - u}{v - u} [g_\lambda(v) - g_\lambda(u)], & v \leq \tau_j, \\ g_\lambda(u) + \phi_j(b_j - u; \lambda), & v > \tau_j, \end{cases} \quad (31)$$

$$T_\lambda(u) = \sum_{j: b_j \geq u} \pi_j [g_\lambda(u) + \phi_j(b_j - u; \lambda)]. \quad (32)$$

An edge accounts only for branches whose target caps it crosses. It therefore carries a contractual allocation value, not an ordinary nearest-neighbor quantization error.

Theorem A.1 (Polynomial price decomposition). *For every λ , some maximizer in (30) has nondecreasing g_λ along its ordered book. Its priced operating value is the sum of (31) over consecutive levels plus (32) at the last level. Set*

$$W_1(i) = T_\lambda(a_i), \quad (33)$$

$$W_r(i) = \max \left\{ T_\lambda(a_i), \max_{\substack{\ell > i \\ g_\lambda(a_\ell) \geq g_\lambda(a_i)}} [E_\lambda(a_i, a_\ell) - \rho_\ell + W_{r-1}(\ell)] \right\}.$$

Then $D(\lambda) = \lambda B + \max_{a_i \leq a_*} [W_m(i) - \rho_i]$. With constant-time evaluation of ϕ_j , computing a maximizing book and this bound takes $O((k + m)N^2)$ arithmetic operations and $O(N^2 + mN)$ storage, allowing both k and m to vary. Rational quadratic data and a rational price give an exact rational algorithm with polynomial bit complexity. General costs use at most $O(kN^2)$ calls to the scalar oracle (29) instead.

Proof. If $\lambda < 0$, strict increase of f makes g_λ strictly increasing. If $\lambda \geq 0$, intermediate service cannot improve priced payoff, and a branch maximizes the interpolant of g_λ on its eligible prefix, truncated at its target cap. Concavity implies that the values along any book increase up to their first maximum and never subsequently increase. All levels after that first maximum may be deleted: for a branch below the maximizing level, either that level is ineligible and all later levels are also ineligible, or later chords cannot improve the concave interpolant before the cap; for a branch above it, no later value beats the maximum. Nonnegative charges make deletion weakly beneficial. The retained book has only admissible arcs.

Consider $b_j \in [u, v)$ on such a book. If $v \leq \tau_j$, the best priced target is b_j on the chord $[u, v]$. For $\lambda \geq 0$ its slope is nonnegative and a singleton tail contributes no positive adjustment. For $\lambda < 0$, a singleton tail contributes at most $-\lambda(b_j - u)$, whereas the chord contributes this amount plus a strictly positive reward increment when $b_j > u$. If $v > \tau_j$, every later level is ineligible; the optimum is a singleton at u followed by the best shortfall in (29). Branches beyond the final level have the same singleton-tail form. These disjoint cap intervals cover all branches because the first level is at most $\min_j b_j$. This proves the decomposition, including equality cases.

The acyclic recurrence chooses either to stop or to append the next admissible level, paying each selected charge once. It enumerates paths implicitly rather than enumerating books. Computing edges costs $O(kN^2)$; the dynamic-programming layers cost $O(mN^2)$. For rational quadratic data, every scalar oracle value uses a bounded number of rational operations, and each path value sums at most $O(km)$ such terms. Numerator and

denominator lengths are polynomial in the input bit length; comparisons and path reconstruction therefore preserve polynomial bit complexity. $\square$

Pricing and constrained shortest-path bounds are established optimization mechanisms (Handler and Zang, 1980). The model-specific result here is that one price makes an otherwise nonseparable shared-book allocation decompose into these cap-crossing edges, despite heterogeneous shortfall costs, branch-specific eligibility, and nonsaturated targets. The theorem does not assert a polynomial algorithm for the unrelaxed joint problem or exchange its maximization with minimization over prices.

A.2 Prefix bounds and checkable global search

A node is a mandatory initial segment $p = (i_1, \dots, i_s)$ of a book; its descendants may append only larger indices. If the priced values along p never decrease, its exact priced completion value is

$$U_p(\lambda) = \lambda B + \sum_{h=1}^{s-1} E_\lambda(a_{i_h}, a_{i_{h+1}}) - \sum_{h=1}^s \rho_{i_h} + W_{m-s+1}(i_s). \quad (34)$$

If a decrease occurs in p , concavity places its first priced maximum within p and every larger new level is useless to the priced allocation. In that case, use the fixed-prefix branch-support value, with every mandatory charge retained. This is again the exact priced maximum over completions; additional nonnegative charges cannot help. The empty prefix uses (30).

Theorem A.2 (Finite search and an anytime global interval). *Fix any nonempty finite set of prices. At each prefix use the smallest corresponding $U_p(\lambda)$, evaluate feasible books by the exact inner allocator, and retain the largest feasible net value L . Partition an unpruned node into its own book, when nonempty, and all one-index extensions. Prune a node only when its upper bound is at most L . This procedure terminates after finitely many prefixes and returns the exact catalog optimum. At any interruption it returns*

$$L \leq \mathcal{V}_{m,\mathcal{A}}^\delta(B; \rho) \leq U = \max\{L, \max_{p \text{ unresolved}} \min_{\lambda} U_p(\lambda)\}. \quad (35)$$

For a uniform mesh of spacing h , Theorem F.1 further yields the continuous interval $[L, U + K_m h]$. The optimization gap $U - L$ and discretization error $K_m h$ are separate and observable.

Proof. The preceding completion argument proves each price bound, so their minimum is valid by weak duality. Every feasible book either equals a nonempty node or has a unique next index. Hence the prescribed children and the node's own book are a disjoint exhaustive partition; book size is at most m . A pruned subtree cannot improve the incumbent. The maximum of unresolved upper bounds and the incumbent therefore bounds every feasible book at every stage. If all subtrees have been resolved, the two bounds coincide. The mesh transport is a separate one-sided inequality between the continuous optimum and the exact catalog optimum, so substituting the catalog upper bound proves the last claim. $\square$

A global witness consists of the feasible winning lottery, every split and its complete child list, upper bounds and prices at pruned or unresolved nodes, and a fixed-book supporting price for each explicitly

evaluated node book. A checker can rebuild price tables and verify the partition without replaying the search order or trusting the incumbent-selection code. This is a stronger object than an inner multiplier or a source hash. Equality of one root price bound with a feasible payoff already certifies global catalog optimality without any branching. Worst-case search is still exponential in the alphabet budget; neither this result nor the mesh transfer is an FPTAS claim.

Additional decomposition contribution. Theorems A.1–A.2 use classical Lagrangian bounds and an acyclic path recurrence. Their structural increment is the exact eligibility-aware cap-crossing reward and prefix-completion dominance for nonsaturated allocation with heterogeneous shortfall costs. The generic search framework is not claimed as new. Proposition F.3 establishes sharpness for uniform catalogs, not a lower bound for adaptive grids. The continuous stationary-face mechanism remains generic; its contractual closed-cell reduction remains the structural contribution.

B Pricing Writable Capacity and Other Implementation Resources

Writable state, read-only constants, evaluation work, and compilation are different resources. A finite menu of implementations t for institution I , with charges depending only on capacity m , can be compared by

$$\min_{I,m,t} \{D_m^I + \lambda_w \lceil \log_2 m \rceil + \lambda_R R_{I,t}(m) + \lambda_T Q_{I,t}(m) + C_{I,t}(m)/H_0 + \kappa_I\}. \quad (36)$$

Here H_0 is the number of repeated deployments over which compilation is amortized, and κ_I is an exogenous protocol charge. This is accounting followed by finite comparison, not an additional structural optimization theorem. Its validity requires the stipulated menu charges to be independent of the particular book at a given capacity. Theorem EC.9.1, in contrast, puts actual additive level charges inside book selection.

With only a writable-bit price, capacity 2^w uses the frontier at $\min(2^w, k)$. For the equal-probability three-cap example, the zero-bit loss is 7/16, the one-bit losses are 1/96 under expected and 1/8 under pathwise participation, and two bits attain full information. Thus a bit price strictly between 1/96 and 1/8 selects one bit under expected participation and two under pathwise participation when all other protocol charges agree. A different exogenous κ_I can change that comparison. This example interprets the proved frontiers; it does not estimate the monetary value of certification or customers' tolerance for risk.

C Exact Continuous Quadratic Design

Fix $s \leq m$ and temporarily allow $0 \leq c_1 \leq \dots \leq c_s \leq 1$. For each branch select one of $2s - 1$ modes. A singleton mode S_ℓ and an adjacent-lottery mode L_ℓ have the following closed descriptions:

$$S_\ell : \quad c_\ell \leq t_j \leq b_j, \quad g_j^S = rc_\ell - \frac{q}{2}c_\ell^2 - \frac{\gamma_j}{2}(t_j - c_\ell)^2; \quad (37)$$

$$L_\ell : \quad c_\ell \leq t_j \leq c_{\ell+1}, \quad t_j \leq b_j, \quad c_{\ell+1} \leq b_j + \delta, \quad g_j^L = rt_j - \frac{q}{2}[(c_\ell + c_{\ell+1})t_j - c_\ell c_{\ell+1}]. \quad (38)$$

For S_ℓ , use $y_j = t_j - c_\ell$ and terminal level c_ℓ . For L_ℓ , use $y_j = 0$ and the adjacent unbiased lottery. At $c_\ell = c_{\ell+1}$ the constraints force $t_j = c_\ell$, and the formula is the singleton reward. Therefore collisions are not discarded as singular probability formulas. The quadratic interpolation identity eliminates the ratio in the lottery probabilities before codewords become optimization variables.

Each joint mode assignment, the ordered-book inequalities, and $\sum_j \pi_j t_j = B$ define a compact polytope. If $\rho(c) = \alpha c^2 + \beta c + \eta$ is nonnegative on $[0, 1]$, its objective

$$Q(c, t) = \sum_j \pi_j g_j(c, t) - \sum_{\ell=1}^s \rho(c_\ell) \quad (39)$$

is a quadratic polynomial, not necessarily concave. Nonnegative charges, rather than signs of their coefficients, are the assumption. For example, a decreasing nonnegative certification charge is allowed.

Lemma C.1 (Finite stationary-face search). *A quadratic function on a nonempty compact polytope attains its maximum at a point that is either a vertex or the unique stationary point on the affine hull of some face with negative-definite restricted Hessian. With rational input data, some maximizing point and its value are rational and can be found by finitely many rational linear solves and feasibility tests.*

Proof. Among all faces whose relative interiors contain a global maximizer, choose one of least dimension and a maximizing point x in its relative interior. The derivative along the face's tangent space vanishes and the restricted Hessian is negative semidefinite. If this restriction has a nonzero null direction z , the quadratic is constant along $x + az$ in the affine hull: both its linear and quadratic terms in a vanish. Compactness makes this line reach a proper face while remaining feasible, contradicting the choice of dimension. The restriction is therefore negative definite unless the tangent space is zero dimensional.

Enumerate independent subsets of the defining inequalities, together with the equality constraints, and solve the stationarity equations on each associated affine space. On the face just identified the bordered stationarity matrix is nonsingular; at a vertex, the tangent space is empty and the same conclusion holds. Test feasibility in every original inequality and retain the largest objective among all feasible candidates. Each retained candidate is feasible, and at least one is a global maximizer. Rational elimination gives the rationality claim. Singular bordered systems may be skipped because the argument guarantees a nonsingular maximizing face of smaller dimension; it does not assume nondegeneracy of the original polytope. $\square$

Theorem C.2 (Exact joint continuous design). *For rational quadratic primitives, rational caps and probabilities, rational $B \in [0, \bar{b}]$ and $\delta \geq 0$, and a rational quadratic charge nonnegative on $[0, 1]$, the maximum of (39) over the closed cells (37)–(38), for every $1 \leq s \leq m$, equals $\mathcal{V}_m^\delta(B; \rho)$. The optimum is attained by a rational book, rational targets, and rational lottery probabilities. Deleting the lottery modes gives the exact deterministic frontier at every promise, with the same charges.*

An exact algorithm enumerates at most $(2s - 1)^k$ cells at size s . After eliminating one target, each cell has dimension at most $s + k - 1$ and at most $s + 1 + 4k$ inequalities. Enumerating its independent active sets

and applying Lemma C.1 gives a finite global algorithm with at most

$$\sum_{s=1}^m (2s-1)^k 2^{s+1+4k} \quad (40)$$

linear systems, each of polynomial dimension and rational encoding length in the input size. The sharper rank-limited count replaces 2^{s+1+4k} by $\sum_{r=0}^{\min\{s+k-1, s+1+4k\}} \binom{s+1+4k}{r}$. A lottery branch may impose four separate inequalities: its two chord bounds, its target cap, and its realization eligibility bound. Neither of the latter two is redundant at a general positive tolerance. This is an exponential reference algorithm, not a polynomial-time claim or an extension of the saturated Monge complexity.

Proof. Theorem 2.1 turns every optimal book and target allocation into singleton or adjacent-lottery branch modes represented in these cells. Conversely every cell point implements a feasible policy by the displayed reconstruction; all constraints, including the exact root promise, hold. If ordered slots collide or some slots are unused, deleting redundant slots cannot increase a nonnegative charge. The resulting smaller book and its canonical branch rules appear among the enumerated sizes. Hence overcounting charges at duplicate slots cannot change the maximum across all sizes. These two directions prove equality with the original optimization problem, not merely a relaxation or a fixed-book upper bound.

There are finitely many compact cells and each has a continuous quadratic objective. Lemma C.1 supplies a rational maximizing candidate on each nonempty cell. Adjacent distinct rational levels reconstruct rational probabilities; coincident levels reconstruct a singleton. The mode count, inequality count, and all-subsets upper bound give (40). Standard determinant bounds for rational linear systems give polynomial encoding length for each candidate. Singleton-only modes exactly represent deterministic terminal assignments and intermediate tiers, proving the deterministic statement. $\square$

The reduction solves the unrestricted continuous outer problem at an interior promise, rather than relating fixed-book allocation as global design. Its contract-specific content is the eligibility-prefix reduction and the closed quadratic elimination of lottery probabilities with no loss at collisions. Rational quadratic optimization and active-set reasoning are classical; polynomial-size witnesses are not a new general complexity principle (Del Pia et al., 2017). The fast saturated Monge theorem remains valuable precisely because the general finite search is exponential.

C.1 Pathwise comparison, repeated caps, and small charges

Theorem C.2 computes $V_m^E(B)$, $V_m^P(B)$, and $V_m^D(B)$ for every rational promise, using respectively all lottery modes at $\delta = 1$, all modes at $\delta = 0$, and singleton modes. No claim of pathwise-deterministic equality away from saturation is needed for this characterization. The complete predecessor companion retains the earlier exact institutional comparison path; the current study reports the separately executed budgeted-design comparisons. The zero-memory-loss interval $B \leq \min b_j$ in Theorem EC.4.1 remains important: a positive institution gap is not expected throughout the feasible promise range.

Proposition C.3 (Pathwise randomization can matter in the interior). *For one branch with $b = 9/10$, $B = 9/20$, $r = 2$, $q = \gamma = 1$, two available symbols, and the nonnegative charge $\rho(c) = 3c(1 - c)$,*

$$\mathcal{V}_2^0(B; \rho) = \frac{171}{400} > \mathcal{V}_2^D(B; \rho) = \frac{9}{160}.$$

The pathwise optimum uses book $(0, 9/10)$, equal draw probabilities, and no intermediate tier. The deterministic optimum uses the single level $9/20$.

Proof. A deterministic policy uses only one level $0 \leq c \leq B$; deleting unused levels weakly lowers its charge. Its net payoff is $-81/800 - (11/20)c + 2c^2$, whose maximum over $[0, B]$ is $9/160$ at B . Every nonsingleton canonical pathwise mode has $0 \leq u \leq B \leq v \leq b$, and net payoff $2B - (u+v)B/2 + uv/2 - 3u(1-u) - 3v(1-v)$. This is convex in (u, v) , so its maximum on that rectangle is at a corner. Checking its four corners, together with the singleton case, gives $(u, v) = (0, 9/10)$ and value $171/400$. Both realizations meet the hard cap and their mean is exactly B . $\square$

This example uses a specified nonnegative, nonmonotone level charge; it is not a counterexample asserted for the uncharged problem. It shows why the saturated pathwise collapse cannot simply be imported into the unified interior design model.

Proposition C.4 (Homogeneous repeated-cap aggregation). *For the randomized institutions in (1), branches with identical caps and identical intermediate-cost functions can be aggregated by adding their probabilities, at every B , δ , and codeword charge. This holds for each fixed book and after global book optimization.*

Proof. Such branches have identical eligible prefixes and the same concave response (2). Replacing their targets by their probability-weighted mean preserves the root promise and does not lower total reward by Jensen's inequality. Replicating an aggregate target gives the converse inequality. Charges depend only on the common book. This argument does not aggregate heterogeneous costs or establish a corresponding deterministic restriction. $\square$

The strict off-cap conclusion is also stable under genuinely selected-level charges. In Proposition D.3, the uncharged two-level book $(1/4, 5/8)$ beats every book of at most two cap levels by $1/1280$. If $0 \leq \rho(c) \leq \eta$ on $[0, 1]$ and $\eta < 1/2560$, evaluating this same off-cap book incurs less than that gap, while a cap-only book cannot improve its uncharged payoff after a nonnegative charge. Thus no charged optimum is cap-only. This statement preserves the off-cap property, not necessarily the numerical location $5/8$.

D The Global Continuous Lottery Codebook

The fixed-codebook formula in Theorem EC.3.1 leaves a continuous design problem: both the terminal service levels and their assignment probabilities must be chosen. We now solve that problem globally for quadratic terminal rewards and heterogeneous quadratic intermediate costs. The solution also determines when imposing participation after, rather than before, the private draw eliminates the value of randomization. These are restricted-alphabet design results, distinct from the exact-implementation alphabet in Theorem EC.4.1.

Retain the three-date renewal architecture of Theorem EC.2.1, including the saturated root promise $\bar{b} = \sum_{j=1}^k \pi_j b_j$, strictly ordered caps $0 < b_1 < \dots < b_k < 1$, and positive probabilities summing to one. Throughout this section,

$$f(c) = rc - \frac{q}{2}c^2, \quad h_j(y) = \frac{\gamma_j}{2}y^2, \quad r \geq q > 0, \quad \gamma_j \geq 0. \quad (41)$$

The parameters and probabilities can differ across branches as indicated. Terminal reward is strictly increasing on the feasible cap range, including when $r = q$. All private information retained until the common terminal vertex must be encoded in the charged symbol. The decoder and public observation convention are unchanged.

D.1 Cap anchors and one continuous terminal level

For $u \leq b \leq v$, quadratic concavity gives the interpolation loss

$$f(b) - \frac{v-b}{v-u}f(u) - \frac{b-u}{v-u}f(v) = \frac{q}{2}(b-u)(v-b), \quad (42)$$

with the endpoint value understood when levels coincide. Thus an interior codeword enters the loss affinely until it crosses a branch cap. The highest codeword is different: reducing it below a branch cap creates an intermediate payment and its associated cost.

Lemma D.1 (Cap-anchored representation). *For every alphabet budget $m \geq 1$, an optimum of (EC.5) exists with the following representation. Its lowest codeword is b_1 ; every codeword except the highest belongs to $\{b_1, \dots, b_k\}$; and the highest belongs to $[b_1, b_k]$. A one-symbol optimum is $\{b_1\}$. The highest codeword need not be a branch cap.*

The proof is in Electronic Companion Section EC.5.4.

The lemma does not discretize the highest level. A cap-only restriction can be strictly suboptimal, as Proposition D.3 shows. It also does not assert that every optimal codebook is anchored; it supplies an optimum with this useful form.

D.2 A finite exact algorithm

For adjacent anchored levels $b_i < b_j$, define

$$E(i, j) = \frac{q}{2} \sum_{h=i}^j \pi_h (b_h - b_i)(b_j - b_h). \quad (43)$$

Let $P(s, j)$ be the minimum interpolation loss on caps through b_j using exactly s anchored levels, starting at b_1 and ending at b_j . Set $P(1, 1) = 0$, $P(1, j) = +\infty$ for $j > 1$, and use the ordered recurrence

$$P(s, j) = \min_{i < j} \{P(s-1, i) + E(i, j)\}, \quad 2 \leq s \leq j. \quad (44)$$

This is a standard shortest-path or segmentation recurrence. The substantive reduction is Lemma D.1, which makes it applicable to a jointly optimized continuous lottery codebook.

Suppose that b_i is the last anchored level and z is the final level. For $\ell \geq i$ and $b_\ell \leq z \leq b_{\ell+1}$, put $b_{k+1} = b_k$ and define

$$\begin{aligned} \Psi_{i\ell}(z) = & \frac{q}{2} \sum_{h=i+1}^{\ell} \pi_h (b_h - b_i)(z - b_h) \\ & + \sum_{h=\ell+1}^k \pi_h \left\{ f(b_h) - f(z) + \frac{\gamma_h}{2} (b_h - z)^2 \right\}. \end{aligned} \quad (45)$$

The first sum is the lottery loss below z ; the second is the terminal-reward and intermediate-payment loss above it. Empty sums are zero. For $\ell < k$ the function is strictly convex in z and has the constrained minimizer

$$z_{i\ell} = \left[\frac{r \sum_{h>\ell} \pi_h + \sum_{h>\ell} \pi_h \gamma_h b_h - \frac{q}{2} \sum_{h=i+1}^{\ell} \pi_h (b_h - b_i)}{q \sum_{h>\ell} \pi_h + \sum_{h>\ell} \pi_h \gamma_h} \right]_{[b_\ell, b_{\ell+1}]}. \quad (46)$$

For $\ell = k$, take $z_{ik} = b_k$. Define the terminal cost

$$T_i = \min_{i \leq \ell \leq k} \Psi_{i\ell}(z_{i\ell}). \quad (47)$$

Theorem D.2 (Global frontier and exact complexity). *Under (41), the optimum with at most m symbols is*

$$D_m^{\text{ran}} = \min \left\{ D_1, \min_{\substack{1 \leq s \leq \min(m-1, k) \\ 1 \leq j \leq k}} \{P(s, j) + T_j\} \right\}, \quad (48)$$

where the inner minimum is empty for $m = 1$. Backtracking produces a globally optimal codebook and the adjacent lotteries implementing it. For ordered inputs and $1 \leq m \leq k$, all frontiers through budget m can be computed in $O(mk^2)$ exact rational operations and $O(mk + k)$ stored rational numbers and indices, including backpointers. No codeword grid is required.

For rational input of total binary length L , a reduced-rational implementation has polynomial Turing complexity. A conservative bound is $O(mk^2 S^3)$ bit operations with $S = O(k[L + \log(k+2)])$. This includes exact cost comparisons, clamping, and fraction reduction. For $m \geq k$ the optimal loss is zero; listing additional budget entries requires only their output cost. These guarantees concern the stated algorithm, not an arbitrary nonlinear optimizer for (EC.5).

The proof is in Electronic Companion Section EC.5.5.

D.3 A continuous optimal highest level

Proposition D.3 (An off-cap terminal level is optimal). *Let $(b_1, b_2, b_3) = (1/4, 1/2, 3/4)$, $(\pi_1, \pi_2, \pi_3) = (7/20, 3/5, 1/20)$, $r = 2$, $q = 1$, and $\gamma_j = 1$. The globally optimal two-symbol expected-participation*

codebook is $(1/4, 5/8)$, and

$$D_2^{\text{ran}} = \frac{23}{1280} < \frac{3}{160} = \min_{\substack{|c| \leq 2 \\ c \subseteq \{b_1, b_2, b_3\}}} D(c) = D_2, \quad (49)$$

where the cap-only minimum is over feasible codebooks and $D(c)$ denotes their expected-participation loss. Thus even optimizing all cap-only lotteries misses the unrestricted optimum.

The proof is in Electronic Companion Section EC.5.6.

D.4 Participation after the private draw

The preceding frontier is appropriate when a renewal review approves an expected continuation before the protocol draws its private symbol. Consider instead an agreement that requires the same cap to hold for every private realization after that review. A draw-specific intermediate tier is allowed; only the terminal public state and the charged symbol reach the terminal decoder. Write $D_m^{\text{pw,ran}}$ for the best randomized loss under these pathwise constraints and the same saturated root promise.

Theorem D.4 (Pathwise randomization has no frontier advantage). *For the general renewal family of Theorem EC.2.1, with strictly increasing concave f and convex nondecreasing h_j , and the observation architecture just specified,*

$$D_m^{\text{pw,ran}} = D_m \quad \text{for every } m \geq 1. \quad (50)$$

Consequently (48) and the deterministic frontier give an exact comparison of the two participation institutions under (41). Their exact-optimum alphabet thresholds are both k when strict concavity makes the k distinct full-information actions unique.

The proof is in Electronic Companion Section EC.5.7.

This institutional result is not an assertion that randomization is generally ineffective in dynamic contracts. It uses the saturated root promise, an exogenous recombination, and the charged-information interface. Under expected participation, Theorem D.2 chooses a less expensive lottery protocol; under pathwise participation, the same service model calls for the deterministic frontier. In either institution an operator can price a symbol budget, add its implementation charge to the computed loss, and optimize the budget without confusing read-only program size with retained private information.

E Monge Costs and Classical Linear Matrix Search

The finite dynamic program still optimizes a continuous highest service level. We first prove the contractual cost inequalities and then verify the classical easy-direction completion with the exact order used by the code. A search routine cannot establish the economic reduction that supplies its entries.

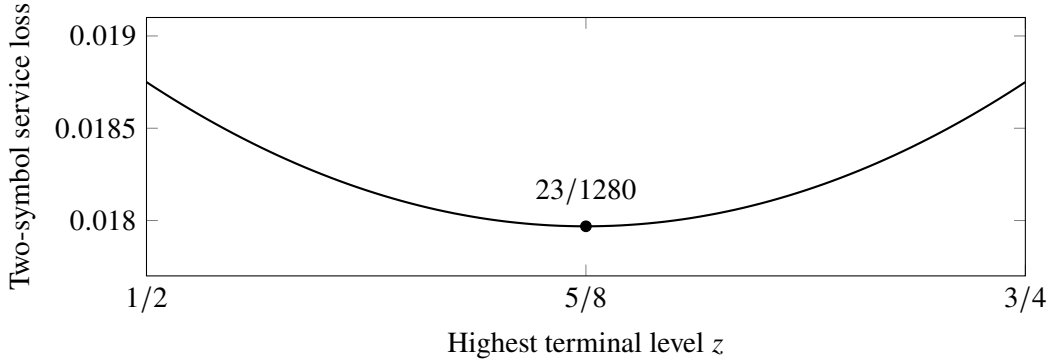


Figure 1: The analytically optimized highest level in the heterogeneous three-branch example. The lower level is $1/4$. The displayed function is $z^2/20 - z/16 + 3/80$ on $[1/2, 3/4]$; its interior minimum is $z = 5/8$, not a branch cap.

E.1 The inequality survives continuous terminal minimization

Write $I_\ell = [b_\ell, b_{\ell+1}]$, including $I_k = \{b_k\}$, and $Q(i, \ell) = \min_{z \in I_\ell} \Psi_{i\ell}(z)$ for $i \leq \ell$. A partial array is Monge here when

$$A(r, c) + A(s, d) \leq A(r, d) + A(s, c), \quad r < s, c < d, \quad (51)$$

whenever all four entries are defined. This fixes the inequality convention before any orientation is named.

Lemma E.1 (Contractual finite-cost inequalities). *The anchored edge costs E and deterministic cell costs C satisfy $A(a, c) + A(b, d) \leq A(a, d) + A(b, c)$ for $a \leq b \leq c \leq d$. The continuously minimized terminal array Q is Monge on its feasible domain. These statements permit arbitrary positive probabilities and heterogeneous nonnegative intermediate curvatures.*

The proof is in Electronic Companion Section EC.5.8.

Lemma E.2 (Previous-layer values are column offsets). *Adding any finite column term v_c preserves the Monge inequality. For fixed finite predecessor values, the arrays $A(j, i) = P(s, i) + E(i, j)$ and $B(j, i) = D(s - 1, i) + C(i + 1, j)$ are therefore Monge wherever all four entries are feasible.*

The proof is in Electronic Companion Section EC.5.9.

E.2 Completion with the implemented strict and tie orders

Give a feasible entry the lexicographic key

$$K(r, c) = (0, A(r, c), c). \quad (52)$$

Finite ties select the smaller column, and distinct columns have distinct keys. The exact total-monotonicity condition used by the implementation is

$$K(r, d) < K(r, c) \implies K(s, d) < K(s, c), \quad r < s, c < d. \quad (53)$$

A strict preference for the right column cannot reverse in a later row. This applies to arbitrary selected rows and columns. Because keys are distinct, a decrease of row-argmin indices in any submatrix would contradict (53).

Lemma E.3 (Suffix-feasible completion). *Let row r have feasible columns $c \geq \ell_r$, with nondecreasing cutoffs ℓ_r . Suppose the finite entries are Monge. Complete every $c < \ell_r$ with $K(r, c) = (1, 0, -c)$. Then (53) holds. A completely infeasible row of a selected submatrix selects its largest selected column.*

The proof is in Electronic Companion Section EC.5.10.

Lemma E.4 (Prefix-feasible completion). *Let row r have feasible columns $c \leq u_r$, with nondecreasing cutoffs u_r . Suppose the finite entries are Monge. Complete every $c > u_r$ with $K(r, c) = (1, 0, c)$. Then (53) holds. A completely infeasible selected row selects its smallest selected column.*

The proof is in Electronic Companion Section EC.5.11.

These lemmas specialize the known favorable completion setting in Section 1; they are not general partial-matrix search theorems. A helper accepting two bounds does not prove total monotonicity for arbitrary interval rows. For example, two zero-cost rows feasible only at columns 2 and 1 have a vacuously Monge finite part but decreasing minima. Uniform infinite padding with ordinary left ties also fails on the required suffix domain: rows with cutoffs 2, 3 restricted to columns 1, 2 select 2, 1. The opposite internal padding orders above are deliberate.

E.3 Reachability and three search interfaces

Lemma E.5 (Contiguous reachable predecessor sets). *For the anchored program, $P(1, j)$ is finite exactly at $j = 1$, and for $s \geq 2$, $P(s, j)$ is finite exactly when $s \leq j \leq k$. For the deterministic program, $D(0, j)$ is finite only at $j = 0$, and $D(s, j)$ is finite exactly when $s \leq j \leq k$ for $s \geq 1$. Each globally reachable predecessor set that can extend to a further endpoint is a contiguous interval or the initial singleton.*

The proof is in Electronic Companion Section EC.5.12.

The following pseudocode uses one-based cap indices and zero for the empty deterministic prefix. “Search” means ordinary full-matrix SMAWK on the supplied key. All prior-layer values are known before a search.

Terminal interface. Rows and columns are $1, \dots, k$. At (i, ℓ) return $(1, 0, -\ell)$ if $\ell < i$; otherwise minimize (45) by (46) and return $(0, Q(i, \ell), \ell)$. Search all rows. Reevaluate each winning cell once to store T_i and its continuous optimizer.

Anchored-prefix interface. To form layer $s+1$, use rows $j = s+1, \dots, k$. Columns are $\{1\}$ for $s = 1$, and $s, \dots, k-1$ otherwise. At (j, i) return $(1, 0, i)$ if $i \geq j$, and $(0, P(s, i) + E(i, j), i)$ otherwise. Search and store values and predecessors. Scan the current reachable endpoints with T_i to update the at-most-budget incumbent; backtrack only its winning codebook.

Deterministic-prefix interface. To form layer s , use rows $j = s, \dots, k$. Columns are $\{0\}$ for $s = 1$, and $s-1, \dots, k-1$ otherwise. Return $(1, 0, i)$ if $i \geq j$, and $(0, D(s-1, i) + C(i+1, j), i)$ otherwise. Search, store predecessors, and backtrack the solution ending at k .

Lemma E.5 justifies every discarded column. The terminal interface has shrinking feasible suffixes; the program interfaces have expanding feasible prefixes. The completion lemmas include infeasible rows in submatrices created internally by SMAWK, although every original searched row has a feasible entry. Padded comparisons do not evaluate an undefined economic cost.

Theorem E.6 (Contractual corollary of classical matrix search). *For ordered rational inputs satisfying (41) and $1 \leq m \leq k$, both expected-participation and deterministic/pathwise frontiers through budget m , including an optimal codebook for every budget, require $O(mk)$ rational operations and comparisons and $O(mk + k)$ stored rational numbers and indices. The highest randomized level remains continuous. With total input length L and $S = O(k[L + \log(k + 2)])$, a conservative Turing bound is $O(mkS^3)$. Sorting unordered distinct caps adds $O(k \log k)$ exact comparisons.*

The proof is in Electronic Companion Section EC.5.13.

The retained divide-and-conquer implementation is a valid $O(mk \log(k + 1))$ alternative using ordinary row-minimum monotonicity. It is an implementation comparator rather than another headline theorem. $O(k)$ exact-rational oracle/comparison work per layer does not imply uniform wall-time superiority for a particular Python implementation.

F Catalog Approximation Without Promise or Risk Slack

Finite catalogs have two different interpretations. An externally certified menu is a restriction of the feasible technology, for which an exact catalog optimum is the relevant answer. Alternatively the operator may choose a numerical catalog to approximate continuous design. The latter interpretation needs a quantitative comparison to the unrestricted optimum; exactness on the catalog alone does not supply it.

Let $\mathcal{A} \subset [0, 1]$ contain zero and one, and let h be its largest consecutive spacing. Assume f is increasing, concave, and L_f -Lipschitz, and each h_j is convex, nondecreasing, and H_j -Lipschitz on $[0, b_j]$. Let the nonnegative charge ρ be L_ρ -Lipschitz on $[0, 1]$. Define

$$K_m = L_f + \sum_j \pi_j H_j + m L_\rho. \quad (54)$$

For the quadratic model, one can take $L_f = r$ and $H_j = \gamma_j b_j$. Neither the root promise nor the risk tolerance is discretized.

Theorem F.1 (Uniform exactly feasible catalog certificate). *For every $B \in [0, \bar{b}]$, every $\delta \geq 0$, and every $m \geq 1$,*

$$0 \leq \mathcal{V}_m^\delta(B; \rho) - \mathcal{V}_{m, \mathcal{A}}^\delta(B; \rho) \leq K_m h. \quad (55)$$

More strongly, every canonical continuous-book policy has a catalog policy with no more symbols that preserves each branch target exactly, satisfies the same realization ceiling exactly, and loses at most $K_m h$ in net payoff. The conclusion includes the pathwise endpoint $\delta = 0$ and requires no strict feasibility margin.

Proof. Round each original level c downward to $a(c) = \max\{a \in \mathcal{A} : a \leq c\}$ and merge collisions. Fix a branch target t and let d be its old eligible top. Every old eligible level remains eligible after

rounding. In particular the new eligible top d' satisfies $d' \geq a(d) \geq d - h$. The old canonical terminal mean is $\mu = \min(t, d)$; coupling its draws to their rounded levels gives a feasible-support lottery of mean $\tilde{\mu} \in [\mu - h, \mu]$ and expected terminal reward at least the old reward minus $L_f h$.

Do not retain those coupled probabilities. Recompute the canonical lottery on the rounded eligible prefix at the *same* target t . Its terminal mean $\mu' = \min(t, d')$ is at least $\tilde{\mu}$: every rounded old draw is at most d' and its mean is at most t . The new interpolant is increasing, and at mean $\tilde{\mu}$ it dominates the coupled lottery reward. Therefore its reward at μ' also dominates that reward. Moreover $\mu' \geq \mu - h$, so its intermediate tier $t - \mu'$ exceeds the old tier by at most h . The branch operating-payoff loss is at most $(L_f + H_j)h$. Theorem 2.1 ensures exact mean and risk feasibility for this reoptimized lottery, including when the risk bound is binding. All t_j are unchanged, so the root promise remains exactly B .

For each original level, $\rho(a(c)) \leq \rho(c) + L_\rho h$. After merging, the total charge is no larger than the sum of these rounded-slot charges because charges are nonnegative. At most m slots were present. Summing the branch bounds and charge bound gives $K_m h$. Apply the construction to an optimizer, or to a sequence approaching the supremum, and use inclusion of catalog policies in continuous policies for the other inequality. $\square$

One-sided rounding is necessary but not sufficient without the probability repair. For example, let $b = t = 1/2$, $\delta = 1/8$, and old support $(1/4, 5/8)$ with upper probability $2/3$. Rounding the lower level to zero while retaining the probabilities reduces the mean to $5/12$. Compensating by $y = 1/12$ produces a maximum obligation $17/24 > 5/8$, violating the risk limit. Instead, the new canonical lottery on $(0, 5/8)$ uses upper probability $4/5$ and $y = 0$. It preserves the target and the ceiling exactly. This is the distinction between unconstrained quantizer rounding and the contractual transport in Theorem F.1.

Corollary F.2 (Certified approximation for a fixed alphabet budget). *For rational quadratic operating primitives and a nonnegative charge whose rational catalog evaluations and Lipschitz bound are supplied, enumerate every feasible catalog book of size at most m and solve its allocation by (4). With $N = |\mathcal{A}|$, this gives an exact catalog optimum in*

$$O\left(\left[\sum_{s=1}^{\min(m, N)} \binom{N}{s}\right] (km + k \log(k + 1))\right) \quad (56)$$

rational arithmetic/evaluation operations. A uniform mesh with denominator $\lceil K_m/\varepsilon \rceil$ (or one when that ceiling is zero) gives additive error at most ε . The algorithm is polynomial in k , K_m/ε , and the bit complexity of the sampled data for fixed m ; it is not claimed to be polynomial in a variable m , nor is this a multiplicative approximation ratio.

Proof. Increasing subsets exhaust the allowed books and the supporting-price algorithm optimizes every inner problem exactly. Take the largest charge-adjusted value and apply Theorem F.1. The mesh includes zero, so every feasible promise is implementable even when it is smaller than the first positive grid point. The bound counts evaluation work; a black-box charge oracle is not assumed to have unit bit complexity unless that is explicitly part of the input model. $\square$

The guarantee is uniform in the two contractual constraints. For any institution pair, subtracting their certified value intervals gives a certified interval for the institutional gap. This permits a global accuracy statement away from saturation without presuming that original cap anchors or Monge order survive. When $B = \bar{b}$, the faster charged path recurrence in Section EC.9 replaces subset enumeration. When the saturated expected problem is uncharged, the continuous Monge algorithm avoids a mesh entirely. Thus the specialized algorithms are faster members of one feasible-policy hierarchy, while (55) connects numerical catalogs to continuous risk-and-charge design.

F.1 The uniform-mesh rate is sharp

The linear bound in Theorem F.1 reflects exact feasibility, not only an insensitive Lipschitz estimate. It cannot be uniformly replaced by a higher-order rate for the declared uniform catalogs.

Proposition F.3 (A fixed-input lower bound). *Take one branch and one symbol, $b = B = 1/2$, $f(c) = c - c^2/2$, $h(y) = y^2/2$, and zero charges. For every risk tolerance and every odd integer $N \geq 3$, the uniform catalog with spacing $h = 1/N$ has loss*

$$\mathcal{V}_1^\delta(B; 0) - \mathcal{V}_{1, \mathcal{A}_N}^\delta(B; 0) = \frac{h}{4} + \frac{h^2}{4}.$$

Thus the worst-case $O(h)$ rate is order-sharp, even for a single fixed input with no opening charges and no randomization.

Proof. With one level, $c \leq B$ and $y = B - c$. Payoff $f(c) - (B - c)^2/2$ is strictly increasing on $[0, B]$. Its continuous maximizer is B , with value $3/8$. For odd N , the largest feasible catalog point is $B - h/2$. Substitution gives the displayed loss. All realizations equal total service B , so risk tolerance plays no role. The sequence of odd N uses the same primitives throughout. $\square$

This is a lower bound for uniform catalogs, not for all adaptive discretizations. Adding the cap $1/2$ removes this particular error. It also does not rule out faster convergence on other fixed instances. The search interval in (35) is therefore useful even when its optimization gap has already vanished: refinement addresses a genuinely different source of error.

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Table 1: Theorem-level contribution map. Prior labels and proofs are retained.

Component	Scope	Provenance
Canonical allocation and continuous design	Exact response; continuous and saturated Monge results	Inherited, Theorems 2.1 and E.6
Saturated charged path	Globally exact because targets are forced	R42 antecedent; Corollary 3.2
Unrestricted price path and prefix search	Polynomial price oracle; finite joint search	R43, Theorems A.1–A.2
Conditional compression and complete target net	Fixed targets; fixed-branch additive joint scheme	R45, Theorems 3.1 and EC.12.2
Catalog-response type reduction	Exact aggregation with prefix-equivalent ceilings	Proposition 4.1; extends homogeneous aggregation
Two-sided box-price path	Exact support on arbitrary target boxes, original budget and charges	New, Theorem 5.1
Adaptive joint interval	Complete box cover; local tightness; positive-accuracy termination	New, Theorem 6.1
Independent certificate checker	Original policy, Bellman inequalities and whole target cover	New implemented verifier

Table 2: Joint design at absolute tolerance 0.001. Counts include every frozen holdout case. Both methods optimize targets and books; neither column is conditional optimization.

Branches	Catalog	Cases	Box completed	Prefix completed	Box seconds	Box max gap
2	9	4	4	4	0.018	0.00057
2	17	4	4	4	0.057	0.00010
4	9	4	4	4	0.028	0.00023
4	17	4	4	4	0.093	0.00094
8	9	4	3	4	0.106	0.01248
8	17	4	2	4	4.103	0.00673
128	33	4	3	4	0.398	0.00229
256	33	2	2	2	0.306	0.00078
16	33	2	2	2	0.850	0.00075

Seconds are medians for box optimization, excluding independent verification. The price-prefix baseline uses the inherited R43 completion bound, not merely an unpriced catalog relaxation. The 128- and 256-branch rows use exactly repeated types; the 128-branch row contains both four- and eight-type cases. Node and time limits, and all individual outcomes, are reported in the companion and machine-readable records.

Table 3: Completed and interrupted target grids versus adaptive joint design on identical finite catalogs. All have $N = 5$ and $m = 2$. A dash means that the grid does not certify the requested accuracy.

Branches	Tolerance	Grid profiles	Grid seconds	Grid certified	Box nodes	Box gap
2	0.100	7	0.005	Yes	1	0.00000
2	0.010	54	0.033	Yes	1	0.00000
2	0.001	527	0.311	Yes	1	0.00000
3	0.100	1000	0.826	–	1	0.00003
3	0.010	1000	0.904	–	1	0.00003
3	0.001	1000	0.908	–	1	0.00003
4	0.100	1000	1.006	–	1	0.01354
4	0.010	1000	1.063	–	3	0.00903
4	0.001	1000	1.086	–	1001	0.00128
5	0.100	1000	1.192	–	1	0.00001
5	0.010	1000	1.163	–	1	0.00001
5	0.001	1000	1.172	–	1	0.00001

The grid has a declared cap of 1,000 evaluated profiles; these are profile-limited comparisons, not claims of time-matched speedup. Adaptive runs use at most 1,001 nodes and an eight-second soft limit. Exact values, objective scales, losses, and runtimes for both algorithms appear in the archived records.

Table 4: Original-budget comparison across six seeds

Charges	Cases	Core exact	Safeguarded exact in sample	Core wins	Greedy wins	Max core loss
None	18	18	18	2	0	0.000000
Nonuniform	18	17	18	8	1	0.028056

Exact means equality with the exhaustive joint catalog reference. Core and greedy wins compare their same-budget values; ties are omitted. The safeguarded policy keeps the better feasible candidate. The charged exception remains in the data. These counts are observations, not global guarantees for a finite target portfolio.

Table 5: Completed same-budget target-net comparisons

Branches	Tolerance	Cases	Profiles	Maximum actual loss
2	1/4	4	28	0.000000
2	1/8	4	50	0.000000
2	1/16	4	96	0.000000
3	1/2	4	47	0.049805
3	1/4	4	164	0.034645

All completed nets meet the requested additive tolerance with the original two-symbol budget and nonuniform charges. Counts aggregate four seeds. An additional interrupted run withholds its global accuracy guarantee.

Table 6: Conditional frontier scaling (targets fixed) with heterogeneous realization ceilings

Seed	Branches	Catalog	Budget	Charges	Selected	Seconds	Gap
104729	32	17	4	No	4	0.036	0.019173
104729	32	17	4	Yes	4	0.036	0.037009
104729	128	33	8	No	8	0.486	0.004410
104729	128	33	8	Yes	5	0.484	0.041534
104729	256	65	8	No	8	3.257	0.004923
104729	256	65	8	Yes	8	3.278	0.025671
205019	32	17	4	No	4	0.038	0.010627
205019	32	17	4	Yes	3	0.038	0.040935
205019	128	33	8	No	8	0.474	0.003661
205019	128	33	8	Yes	5	0.478	0.042000
205019	256	65	8	No	8	3.303	0.004099
205019	256	65	8	Yes	7	3.316	0.023976

Seconds are medians of three repeated executions on the same input. Gap is the valid same-budget Jensen upper bound minus the ideal-target conditional policy value, not an observed error to an unknown global optimum. Full timing arrays, stored-entry bit lengths and cumulative process resident-memory records are archived.

Table 7: Adaptive candidates at an unchanged installed-symbol budget

Case	Candidate pool	Candidates	Symbols	Exact-reference loss
interior-ideal	uniform	9	2	0.044030
interior-ideal	uniform	65	2	0.002991
interior-ideal	oracle pool	4	2	0.000000
off-cap-top	uniform	9	2	0.000000
off-cap-top	Adaptive	5	2	0.000781
off-cap-top	Adaptive	6	2	0.000000
Sat. 104729	uniform	9	3	0.006507
Sat. 104729	Adaptive	9	3	0.000000
Sat. 205019	uniform	9	3	0.001824
Sat. 205019	Adaptive	9	3	0.000000

Every row has an exact continuous reference in the stated Jensen-attaining or saturated expected-participation regime. The interior oracle pool explicitly inserts ideal targets and is a sanity check. Candidate-pool size is not installed memory. Adaptive midpoint selection is a heuristic; catalog optimization at the saturated target is exact. The complete 22-row path is in the companion and JSON.